NARESH RAMESH KHANDERAY

Data Analyst | Data Engineer | AI/ML Engineer | Python Developer

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PROFESSIONAL SUMMARY

Results-driven Data Analyst and AI/ML Engineer with hands-on expertise in building end-to-end machine learning pipelines, deploying scalable REST APIs, and delivering actionable BI dashboards. Proficient in Python, SQL, PySpark, and AWS cloud services. Experienced with Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Generative AI development using LangChain and OpenAI API. Completing a PG Diploma in Big Data Analytics at C-DAC Mumbai (Aug 2025). Passionate about transforming raw data into measurable business impact through scalable, AI-driven solutions.

PROFESSIONAL EXPERIENCE

AI Consultant — House of EdTech 25 May – 6 Sept

- Design and implement AI-driven solutions for ed-tech products, including LLM-powered learning assistants, content generation tools, and personalised recommendation systems.
- Build and optimise Retrieval-Augmented Generation (RAG) pipelines using LangChain and OpenAI APIs to deliver accurate, context-aware responses for student and educator-facing applications.
- Collaborate with product and engineering teams to integrate AI/ML models into existing platforms, translating business requirements into scalable technical solutions.
- Advise on best practices for prompt engineering, model evaluation, and data pipeline design to improve the quality, reliability, and performance of AI features.

TECHNICAL SKILLS

Programming Languages	Python, SQL, R
ML / AI & GenAI	Scikit-learn, XGBoost, TensorFlow, Keras, NLP (NLTK, BERT, Hugging Face), LangChain, RAG, Prompt Engineering, OpenAI API, LLM Fine-tuning, Vector Databases
Data & Analytics	Pandas, NumPy, EDA, Feature Engineering, Statistical Analysis, Hypothesis Testing, A/B Testing, Time Series Analysis, SMOTE
Data Engineering	ETL Pipelines, Data Warehousing, Database Design, SQL Optimisation, Data Modelling, Apache Kafka
Big Data	Apache Spark, PySpark, SparkSQL, Hadoop/HDFS, Hive, AWS EMR
Cloud & DevOps	AWS (EC2, S3, Lambda, SageMaker, EMR), Docker, Flask, REST APIs, Git, Linux, CI/CD
Visualisation & BI	Power BI, Tableau, Matplotlib, Seaborn, Excel (Advanced), DAX, Power Query

KEY PROJECTS

Olist E-Commerce Real-Time Intelligence Platform

Tech Stack: Python · PySpark · SparkSQL · AWS (S3, EC2, Lambda, SageMaker, EMR) · NLP (BERT, Hugging Face) · Flask · Power BI · Tableau · MySQL · Docker · GitHub

- Engineered an end-to-end retail analytics pipeline on the Brazilian Olist dataset (100K+ orders) using PySpark on AWS EMR, enabling large-scale distributed data processing and reducing batch processing time by 40%.
- Fine-tuned a BERT-based sentiment classifier on 40,000+ customer reviews achieving 88% F1-score; surfaced product quality issues that directly informed procurement and vendor management decisions.
- Deployed a Flask REST API on AWS EC2 serving real-time churn prediction and delivery-risk scores integrated with Power BI executive dashboards for C-suite visibility.
- Performed geospatial analysis to identify top 3 delivery bottleneck states; data-driven recommendations projected a 15% reduction in late deliveries and improved customer satisfaction scores.
- Implemented automated data quality checks and monitoring using AWS Lambda triggers, ensuring pipeline reliability and data integrity across the entire workflow.

Retail Customer Churn Prediction System

Tech Stack: Python · Scikit-learn · XGBoost · Pandas · NumPy · Flask · Docker · AWS (EC2, S3) · Power BI · SMOTE

- Processed and feature-engineered a 50,000-record retail/telecom dataset; resolved class imbalance using SMOTE, improving minority-class recall by 23% over baseline.
- Benchmarked Logistic Regression, Random Forest, Gradient Boosting, and XGBoost; final XGBoost model achieved ~85% accuracy and 0.82 AUC-ROC with full hyperparameter tuning via GridSearchCV.
- Containerised the Flask prediction API with Docker and hosted on AWS EC2/S3, delivering scalable, low-latency inference with 99.5% uptime in a simulated production environment.
- Designed a Power BI churn KPI dashboard with DAX measures and drill-through reports, enabling the business team to prioritise retention outreach for high-risk customer segments — targeting 20% reduction in churn.

Intelligent Document Q&A System — RAG + LLM (Generative AI)

Tech Stack: Python · LangChain · OpenAI API · FAISS (Vector DB) · Hugging Face Embeddings · Flask · Streamlit · AWS S3 · PyPDF2 · GitHub

- Built a production-ready Retrieval-Augmented Generation (RAG) pipeline that ingests PDF/text documents, chunks and embeds content using Hugging Face sentence transformers, and stores vectors in FAISS for semantic search.
- Integrated OpenAI GPT API with LangChain's ConversationalRetrievalChain to deliver context-aware, multi-turn Q&A with memory, reducing hallucinations by grounding responses in retrieved document chunks.
- Developed a Streamlit front-end and Flask REST API backend enabling users to upload documents and query them in real time; deployed on AWS EC2 with S3 for document storage.
- Implemented prompt engineering techniques including few-shot prompting and system message tuning to optimise response quality, relevance, and tone across diverse document types.
- Evaluated system performance using RAGAS framework (faithfulness, answer relevancy, context precision), achieving scores above 0.85 across all metrics on a custom evaluation dataset.

EDUCATION

Post Graduate Diploma in Big Data Analytics (PG-DBDA) — C-DAC, Mumbai Feb 2025 – Aug 2025
Intensive program: Python, SQL, R, EDA, ML, Deep Learning, Hadoop, Spark, PySpark, Hive, Power BI, Tableau, Cloud, Real-World Projects

Master of Science (M.Sc.) — Computer Science — Z.B. Patil College, Dhule | North Maharashtra University 2022 – 2024
Bachelor of Science (B.Sc.) — Computer Science — Z.B. Patil College, Dhule | North Maharashtra University 2019 – 2022

CERTIFICATIONS & COURSES

- AWS Cloud Practitioner Essentials — Amazon Web Services
- Python for Data Science & Machine Learning — Udemy / Online
- PG-DBDA Program — C-DAC Mumbai: Python, SQL, R, Machine Learning, Deep Learning, Big Data (Hadoop, Spark), Power BI, Tableau, Cloud Fundamentals

PERSONAL INFORMATION

Date of Birth: 22 September 2001 Nationality: Indian Gender: Male Languages: English, Marathi, Hindi

I hereby declare that the above information is true and correct to the best of my knowledge.
Naresh Ramesh Khanderay